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Methodological Project Report

Selective Inference for Hierarchical Clustering

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1 Introduction

A DIRE:

1. tester si moyennes egal est tres frequent
2. mais on peut pas simplement faire des t-test parce que data sont déjà utilisées par algo clustering
3. donc faut trouver d'autres solutions -> selective inference
4. c'est le but de l'article qu'on étudie qui propose une solution de selective infenrece post clustering

In general, clustering methods do not provide theoretical guarantees regarding the partitions they generate in standard scenarios. To ensure that the results provide guarantees on the actual differences between groups, it is very common to perform tests for equality of means between the different clusters.

However, employing a standard z-test or Wald test results in an inflation of the type I error rate [4]. This arises from the fact that the null hypothesis is constructed adaptively, it is defined only after observing the data and is conditioned on the outcome of the clustering algorithm.

This issue is analogous to overfitting in supervised learning frameworks. When the same dataset is utilized for both model training and evaluation, the resulting model often fails to generalize, capturing noise rather than underlying patterns. In the present context, performing statistical inference without accounting for the variability induced by the clustering procedure results in a failure to maintain the type I error rate.

To address this issue, Gao et al. developed a valid test for the difference in means between two clusters estimated via an hierarchical agglomerative clustering algorithm.

2 Classical versus selective testing

2.1 Hypothesis testing and *p-values*

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $(E, \mathcal{E})$ a topological space and $\mathcal{T}$ the σ-algebra generated by $\mathcal{E}$. A *random variable* is a measurable function $X : (\Omega, \mathcal{F}) \to (E, \mathcal{T})$. The *(probability) distribution* of X is the mapping $P : \mathcal{T} \to [0, 1]$ such that $P(O) = (\mathbb{P} \circ X^{-1})(O)$ for all $O \in \mathcal{T}$. We say that P is *supported* on E and denote by $\mathcal{M}_1(E)$ the set of all probability distributions supported on $\mathcal{E}$. From now on, we will set $E = \mathbb{R}$ and simply write $\mathcal{M}(\mathbb{R}) = \mathcal{M}$.

Definition 1 (Hypothesis, Definition 1.1 in [5])**.** *A hypothesis $\mathcal{H}_0$ is a set of probability distributions in $\mathcal{M}$. A hypothesis is simple if it is a singleton, such as $\{P\}$ or $\{Q\}$, and composite otherwise. The complementary set $\mathcal{M} \setminus \mathcal{H}_0$ is called the alternative hypothesis of $\mathcal{H}_0$.*

Definition 2 (Test)**.** *A test for $\mathcal{H}_0$ is a binary partition of the sample space, defined by a mapping:*

$$\begin{aligned} \pi_{\mathcal{H}_0} : \Omega &\to \{0,1\} \\ \omega &\mapsto \pi_{\mathcal{H}_0}(\omega). \end{aligned} \tag{1}$$

For $\omega \in \Omega$, we say that the test rejects $\mathcal{H}$, based on ω if $\pi_{\mathcal{H}_0}(\omega) = 1$, and does not reject $\mathcal{H}_0$ based on ω if $\pi_{\mathcal{H}_0}(\omega) = 0$.

In what follows, we will adopt the more standard notation:

$$\mathbb{P}\left(\text{Reject } \mathcal{H}_0\right) = \mathbb{P}\big(\{\omega \in \Omega \,:\, \pi_{\mathcal{H}_0}(\omega) = 1\}\big),$$

making the dependence on π implicit.

Definition 3 (Type I error)**.** *We say that a test controls the type I error at level α if*

$$\mathbb{P}(\text{Reject } \mathcal{H}_0) \leq \alpha, \quad \textit{for } \alpha \in (0,1).$$

We say that it controls the type I error exactly at level α if

$$\mathbb{P}(\text{Reject } \mathcal{H}_0) = \alpha, \quad \textit{for } \alpha \in (0,1).$$

Definition 4 (Stochastic dominance)**.** *Let X and Y be real-valued random variables. We say that Y stochastically dominates X, and write*

$$X \preceq_{\text{st}} Y,$$

if

$$\mathbb{P}(X \leq u) \geq \mathbb{P}(Y \leq u), \quad \forall\, u \in \mathbb{R}. \tag{2}$$

Definition 5 (Super-uniform random variable)**.** *Let X be a real-valued random variable. We say that X is super-uniform on [0,1], and write $X \sim \mathcal{SU}(0,1)$, if X stochastically dominates a uniform random variable on $[0,1]$, that is, if*

$$\mathbb{P}(X \leq u) \leq u, \quad \forall\, u \in [0,1]. \tag{3}$$

Proposition 1. *Let X and Y be real-valued random variables, with cumulative distribution functions F_X and F_Y, respectively. If $X \preceq_{\text{st}} Y$, then $F_X(Y) \sim \mathcal{SU}(0,1)$.*

Remark 1. *If $X = Y$, then $F_X(X)$ is super-uniform on $[0,1]$. Moreover, if X has a continuous distribution function F_X, then*

$$F_X(X) \sim \mathcal{U}(0,1). \tag{4}$$

Definition 6 (p-value, Definition 1.1 in [5])**.** *Let $\mathcal{H}_0$ be a hypothesis. A p-value for $\mathcal{H}_0$ is a super-uniform random variable under $\mathcal{H}_0$.*

We often build tests using p-values as $\pi = \mathbb{1}\{p \leq \alpha\}$.

Proposition 2. *Let $\mathcal{H}_0$ be a hypothesis and p a p-value for $\mathcal{H}_0$. Then, the test $\pi = \mathbb{1}\{p \leq \alpha\}$ controls the type I error at level α, for $\alpha \in (0, 1)$.*

Definition 7 (Test statistic)**.** *A test statistic is a measurable function $T : \Omega \to \mathbb{R}$.*

From now on, we will consider the case of unilateral tests. In this setting, to define a test based on the information given by a real-valued random variable X, p-values are often built in the form:

$$p(x) = \mathbb{P}_{\mathcal{H}_\prime}\big(T(X) \geq T(x)\big), \tag{5}$$

where T is a test statistic and x a realization of X.

The p-value (5) follows a uniform distribution under $\mathcal{H}_0$. However, we have defined p-values as being super-uniform random variables under the null hypothesis, which corresponds to a more general family of distributions. To adapt the classical form (5) to that setting, we introduce the following proposition.

Proposition 3. *Let T and T' be two test statistics. Let X be a random variable and x a realization of X. Define*

$$p(x) = \mathbb{P}_{\mathcal{H}_0}\big(T'(X) \geq T(x)\big). \tag{6}$$

If

$$T'(X) \preceq_{\text{st}} T(X) \quad \textit{under } \mathcal{H}_0, \tag{7}$$

then p is a p-value for $\mathcal{H}_0$.

Remark 2. *If $T' = T$ and the cumulative distribution function F_T of $T(X)$ is continuous under $\mathcal{H}_0$, then*

$$p \overset{\mathcal{H}_0}{\sim} \mathcal{U}(0, 1).$$

In the classical setting, the null hypothesis is independent of the data. However, in many practical applications $\mathcal{H}_0$ is chosen *after seeing the data*. In this framework, the classical testing approaches built for type I error control are unsuitable. Instead, statistical guarantees need to be provided via *selective inference*. In particular, the theory of statistical testing of data-driven null hypotheses is known as *selective testing* [3]. The next section provides some examples that motivate this framework.

2.2 Examples motivating selective testing

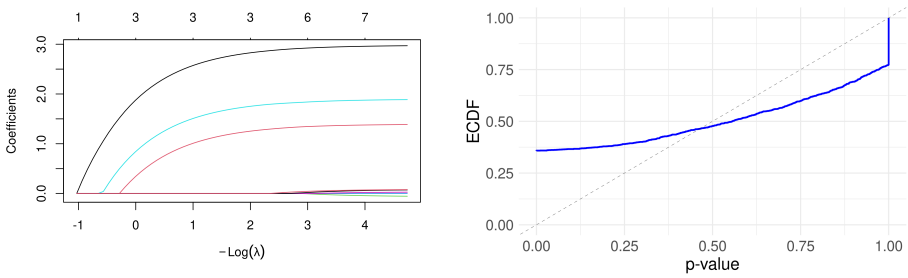
2.2.1 Lasso

To determine whether an explanatory variable helps explain a response variable through a linear regression model, a common practice is to test whether its associated coefficient is significantly different from zero. In the context of classical linear regression, this falls within the framework of non-selective inference, because all coeficients are fixed *a priori*. In contrast, Lasso regression uses an ℓ_1-penalty to perform variable selection [6]. As only the coefficients selected by the Lasso can be tested, the null hypothesis depends on the outcome of the Lasso regression, and is therefore data-driven. This corresponds to a setting of selective inference, where the classical control of type I error fails.

To illustrate the unsuitability of non-selective inference, consider a centered Gaussian vector $X = (X_1, \ldots, X_8)$ and set the model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1), \quad \beta = (3, 1.5, 0, 0, 2, 0, 0, 0)^\top.$$

We generated $n = 100$ independent realizations of (X, Y) and implemented the Lasso algorithm, obtaining the regularization path presented in Figure 1(a). Then, we tested whether a randomly chosen coefficient selected by Lasso regression and among the trully null coefficients, equals 0. After repeating this pipeline $M = 2000$ times, we obtained the empirical p-value distribution depicted in Figure 1(b). We clearly see that the p-values are not super-uniform, motivating the use of a selective testing approach instead of naive testing after selection.



(a) (b)

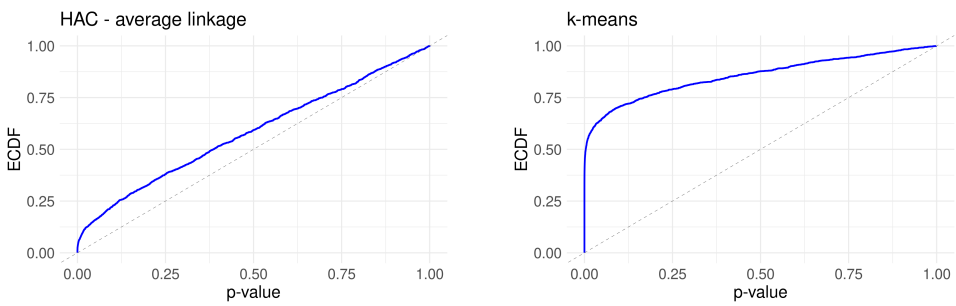
Figure 1: (a) Regularization path of the coefficients obtained using the Lasso algorithm with a single realization of (X, Y). Some coefficients are zero, and three converge toward the theoretical values (3,1.5,2). (b) Empirical cumulative distribution function (ECDF) of the p-values obtained from tesing wheter a randomly chosen coefficient is null after selecting the coefficient by a Lasso regression. The ECDF was computed from $M = 2000$ simulations.

2.2.2 Publication bias

In many areas of research, scientists tend to test for signficance only when the associated effect is found to be substantial. In other words, testing is performed after a selection process that filters out small effects. In that setting, controlling type I error at a fixed level α yields inflated false positive rates when considering the ensemble of all published studies. If $Y_i \sim \mathcal{N}(\mu_i, 1)$ represents the effect size in a scientific study, whose significance is tested only if $|Y_i| > 1$, a naive level α test $H_{0,i} : \mu_i = 0$ is invalid. Indeed, Fithian [3] demonstrates that the false positive rate among true nulls reaches approximately 0.16, far exceeding the nominal 0.05 level. Valid inference requires thresholding $|Y_i|$ at 2.41 rather than 1.96, the 0.95 quantile of the standard normal, imposing a more stringent criterion.

2.2.3 Clustering

Another remarkable example of selective inference appears when evaluating the performance of clustering algorithms by testing for the equality of cluster means. To illustrate the need of using appropriate tests in the context of selective inference, we simulated $n = 100$ samples of a five-dimensional standard Gaussian random vector. Each sample was classified into $K = 3$ groups using hierarchical agglomerative clustering and k-means. Then, the equality of cluster means was tested using a classical t-test, for two randomly selected clusters. If the test controlled the type I error in this setting, the resulting p-value would be super-uniformly distributed under the null hypothesis. However, the 'clustering + post-selection testing' procedure leads to a deviation from super-uniformity, as shown in Figure 2.



(a) (b)

Figure 2: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for (a) hierarchical agglomerative clustering (HAC) and (b) k-means algorithms. The distribution functions were computed using $M = 2000$ simulations from a multivariate centered normal distribution.

2.3 Addressing selective testing

Since the type I error is no longer controlled, the definition of alternative error risks is required in this setting. Accordingly, we present three methods as described by Yoav Benjamini [1].

2.3.1 Simultaneous Inference

This approach controls the family-wise error rate across all hypotheses:

$$\mathbb{P}(\text{At least 1 false positive among all hypotheses}) \leq \alpha.$$

This strategy proves highly conservative, ensuring that for every possible set of hypotheses, the probability of at least one false positive remains below α. To date, no post-clustering inference approaches based on this type of control have been proposed.

2.3.2 Sample splitting

This approach consists in splitting the dataset into a training set X and a test set Y. The training set X is used to choose the null hypothesis $\mathcal{H}_0(X)$. Then, inference is performed on the test set Y.

Although relatively simple to implement, this method raises several issues. First, statistical guarantees on the tests hold only if X and Y are independent, which is rarely the case in practice. In addition, comparing cluster means on the observations in Y requires assigning each test point to one of the clusters obtained from the clustering performed on X, a step that compromises validity, as it reuses the dataset X for two separate steps. As discussed in [4], this strategy does not yield valid post-clustering inference in general.

2.3.3 Conditional Inference

This approach constitutes the most extensively studied framework for post-clustering inference. It controls the false positive rate conditional on hypothesis selection:

$$\mathbb{P}(\text{Reject } \mathcal{H}_0(X) \mid \mathcal{H}_0 \text{ selected}) \leq \alpha. \tag{8}$$

In the remainder of the article, we employ this method to develop a statistical procedure for testing equality of cluster means following a clustering algorithm.

3 Post-clustering inference

The article under study [4] was the first one to propose a feasible solution to post-clustering inference. Here, we will focus on the setting and approach introduced by the authors.

3.1 Gao *et al.*'s approach

Let $\mathbf{X} = (X_{ij})_{ij} \in \mathbb{R}^{n\times p}$ be the design matrix. A cluster is an element of a partition of the samples. We note C a clustering algorithm and $C_1 \in C(\mathbf{X})$ a cluster obtained by the algorithm C on $\mathbf{X}$. We note $\boldsymbol{\mu} = (\mu_{ij})_{ij}$ where $\mu_{ij} = \mathbb{E}[X_{ij}]$and, for a subset G of $\{1, \dots, n\}$, we write

$$\bar{\mu}_G = \frac{1}{|G|}\sum_{i\in G}\mu_i \in \mathbb{R}^p \quad \text{and} \quad \bar{X}_G = \frac{1}{|G|}\sum_{i\in G} X_i \in \mathbb{R}^p.$$

The goal is to test for the equality of cluster means, that is, testing the following null hypothesis: Use the same notation $\mathcal{H}_0^{\{C_1,C_2\}}$ for the null everywhere.

$$\mathcal{H}_0^{\{C_1,C_2\}} : \bar{\mu}_{C_1} = \bar{\mu}_{C_2}, \tag{9}$$

for a pair of clusters C_1, C_2. For concision and clarity, we introduce the n-dimensional vector

$$\nu = (\nu_i)_{i=1}^n, \qquad \nu_i = \begin{cases} \dfrac{1}{|C_1|}, & \text{if } i \in C_1, \\ -\dfrac{1}{|C_2|}, & \text{if } i \in C_2, \\ 0, & \text{otherwise,} \end{cases}$$

such that the population (resp. empirical) difference of cluster means can be compactely rewritten as:

$$\begin{cases} \bar{\mu}_{C_1} - \bar{\mu}_{C_2} = \nu^T \boldsymbol{\mu} \\ \bar{X}_{C_1} - \bar{X}_{C_2} = \nu^T \mathbf{X}. \end{cases}$$

Thus, we can rewrite the null hypothesis as follows:

$$\mathcal{H}_0 : \bar{\mu}_{C_1} = \bar{\mu}_{C_2} \iff \nu^T \boldsymbol{\mu} = 0. \tag{10}$$

From now on, we adopt the conditional inference framework, described in Section 2.3.3. In this setting, controlling the risk (8) is equivalent to controlling the so-called *selective type I error for clustering*, defined below.

Definition 8 (Selective type I error for clustering)**.** *We say that a test controls the selective type I error for clustering at leval α if*

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left(\textit{Reject } \mathcal{H}_0^{\{C_1,C_2\}} \mid C_1, C_2 \in C(\mathbf{X})\right) \leq \alpha, \quad \alpha \in (0,1). \tag{11}$$

We say that it controls exactly the selective type I error for clustering at level α if

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left(\textit{Reject } \mathcal{H}_0^{\{C_1,C_2\}} \mid C_1, C_2 \in C(\mathbf{X})\right) = \alpha, \quad \alpha \in (0,1). \tag{12}$$

To define a p-value that controls the selective type I error, model assumptions need to be imposed on $\mathbf{X}$. The authors in [4] adopt the following matrix normal model:

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbb{I}_n, \sigma^2\mathbb{I}_p), \tag{13}$$

which means that the lines of $\mathbf{X}$ are independent p-dimensional random vectors distributed as $\mathcal{N}_p(\mu_i, \sigma^2\mathbb{I}_p)$, for every line $i \in \{1, \dots, n\}$. Note that this model imposes an independence assumption between the features (columns) of $\mathbf{X}$.

To run a proper test, we need to control the selective type I error at level α. Ideally, we would like to define a p-value as follows:

$$p_{ideal}(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\big(T(\mathbf{X}) \geq T(\mathbf{x}) \mid C_1, C_2 \in C(\mathbf{X})\big),$$

with T being a test statistic. In [4], Gao *et al.* defined the test statistic as $T(\mathbf{X}) = ||\nu^\top \mathbf{X}||_2$, because its distribution under the null hypothesis can be straightforwardly derived. Indeed, noting that:

$$\nu^\top \mathbf{X} \sim \mathcal{MN}_{1\times p}\big(\nu^\top \mu,\ \nu\nu^\top,\ \sigma^2 I_p\big) = \mathcal{N}_p\big(\nu^\top \mu,\ \|\nu\|_2^2 \sigma^2 I_p\big),$$

we have, under $\mathcal{H}_0^{\{C_1,C_2\}}$, that

$$||\nu^\top \mathbf{X}||_2 \sim ||\nu||_2 \sigma \chi_p. \tag{14}$$

However, p_{ideal} cannot be evaluated in practice as it depends on unknown parameters [4]. To address this issue, the authors in [4] propose to add technical events to the conditioning set, considering the following quantity:

$$p(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(||\nu^T \mathbf{X}||_2 \geq ||\nu^T \mathbf{x}||_2 \mid C_1, C_2 \in \mathcal{C}(\mathbf{X}), Ev(\mathbf{X})\Big), \tag{15}$$

as a p-value for (10), with $Ev(\mathbf{X})$ being the additional events. Maybe it is better to change the notation $E(\mathbf{X})$ to something else to avoid the confusion with the expectation of $\mathbf{X}$. You can choose another notation for that quantity. To justify the choice of $Ev(\mathbf{X})$, we follow the steps in [4]. First, we decompose the data matrix as follows:

$$\mathbf{X} = \pi_\nu^\perp \mathbf{X} + (\mathbb{I}_n - \pi_\nu^\perp)\mathbf{X}, \tag{16}$$

where $\pi_\nu^\perp$ is the orthogonal projection matrix onto the subspace orthogonal to ν, defined as:

$$\pi_\nu^\perp = \mathbb{I}_n - \frac{\nu\nu^\top}{||\nu||_2^2} \in \mathbb{R}^{n\times n}. \tag{17}$$

Then, we can write

$$\begin{aligned}\mathbf{X} &= \pi_\nu^\perp \mathbf{X} + \frac{\nu\nu^T}{||\nu||_2^2}\mathbf{X} \\ &= \pi_\nu^\perp \mathbf{X} + \frac{\nu}{||\nu||_2^2}\frac{\nu^T\mathbf{X}}{||\nu^T\mathbf{X}||_2}||\nu^T\mathbf{X}||_2 \\ &= \pi_\nu^\perp \mathbf{X} + \frac{\nu}{||\nu||_2^2}\,\mathrm{dir}(\nu^T\mathbf{X})||\nu^T\mathbf{X}||_2,\end{aligned} \tag{18}$$

where dir denotes the direction of a vector, that is,

$$\mathrm{dir}(z) = \frac{z}{||z||_2}, z \in \mathbb{R}^p. \tag{19}$$

Since $\mathrm{dir}(\nu^T\mathbf{X})$ and $\pi_\nu^\perp\mathbf{X}$ are unknown, we condition on their values for a realization $\mathbf{x}$ of $\mathbf{X}$, defining the additional events as:

$$Ev(\mathbf{X}) = \left\{\pi_\nu^\perp\mathbf{X} = \pi_\nu^\perp\mathbf{x},\ \mathrm{dir}(\nu^T\mathbf{X}) = \mathrm{dir}(\nu^T\mathbf{x})\right\}.$$

Plugging the previous equation in (15) we obtain the following expression for the p-value:

$$\begin{aligned}p(x) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(||\nu^\top\mathbf{X}||_2 \geq ||\nu^\top x||_2 \,\Big|\, C_1, C_2 \in \mathcal{C}(\mathbf{X}), \\ \pi_\nu^\perp\mathbf{X} = \pi_\nu^\perp\mathbf{x},\ \mathrm{dir}(\nu^\top\mathbf{X}) = \mathrm{dir}(\nu^\top\mathbf{x})\Big).\end{aligned} \tag{20}$$

Now that we have defined the model, the hypothesis to be tested, and the corresponding p-value, we will show that this p-value can be computed under an analitically tractable form and that it properly controls the selective type I error. For this, we rely on the following technical lemma, whose proof is presented in Appendix A.

Lemma 1. *Under the null hypothesis, the following statements hold:*

(i) $\|\nu^\top\mathbf{X}\|_2 \perp\!\!\!\perp \pi_\nu^\perp\mathbf{X}$,

(ii) $\|\nu^\top\mathbf{X}\|_2 \perp\!\!\!\perp \mathrm{dir}(\nu^\top\mathbf{X})$.

Following from the previous result, we can rewrite the selective p-value as:

$$\begin{aligned}p(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(||\nu^\top\mathbf{X}||_2 \geq ||\nu^\top\mathbf{x}||_2 \ | \\ C_1, C_2 \in \mathcal{C}(\pi_\nu^\perp\mathbf{X} + \frac{\nu}{||\nu||_2^2}\,\mathrm{dir}(\nu^\top\mathbf{x})||\nu^\top\mathbf{x}||_2)\Big)\end{aligned} \tag{21}$$

Rewrite the previous equation in two lines.

Letting $\phi = ||\nu^\top \mathbf{X}||_2$ and $x'(\phi) = \pi_\nu^\perp \mathbf{x} + \frac{\nu}{||\nu||_2^2} \operatorname{dir}(\nu^\top \mathbf{x})\phi$ we can write

$$\begin{aligned} p(\mathbf{x} &= \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(\phi \geq ||\nu^\top \mathbf{x}||_2 \mid C_1, C_2 \in \mathcal{C}\big(\mathbf{x}'(\phi)\big)\Big) \\ &= \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(\phi \geq ||\nu^\top \mathbf{x}||_2 \mid \phi \in \mathcal{S}\big(\mathbf{x}; \{C_1, C_2\}\big)\Big), \end{aligned} \tag{22}$$

where

$$\mathcal{S}\big(\mathbf{x}, \{C_1, C_2\}\big) = \Big\{\phi \geq 0 : C_1, C_2 \in \mathcal{C}\big(\mathbf{x}'(\phi)\big)\Big\}. \tag{23}$$

As, under the null, we have that $\phi \sim ||\nu||_2 \sigma \chi_p$, we can finally write:

$$p(\mathbf{x}) = 1 - \mathbb{F}_p\Big(\nu^\top \mathbf{x}; ||\nu||_2 \sigma; \mathcal{S}\big(\mathbf{x}; \{C_1, C_2\}\big)\Big), \tag{24}$$

where $\mathbb{F}_p(t; c; \mathcal{S})$ denotes the cumulative distribution function of a $c \cdot \chi_p$ random variable truncated to the set $\mathcal{S}$. This proves that the p-value (20) can be written in the analitically tractable form (24). This corresponds to the first statement in [4, Theorem 1], that we present below.

Theorem 1 (Theorem 1 in [4])**.** *For any realization* $\mathbf{x}$ *of* $\mathbf{X} \sim \mathcal{MN}_{n\times p}(\mu, \mathbb{I}_n, \sigma^2 \mathbb{I}_p)$*, and for any non-overlapping groups of observations* $C_1, C_2 \subseteq \{1, ..., n\}$*, Maybe try to rephrase so that the equation doesn't fall outside the text limits.*

$$p(\mathbf{x}) = 1 - \mathbb{F}\big(\nu^\top \mathbf{x};\ ||\nu||_2 \sigma;\ \mathcal{S}\big(\mathbf{x}; \{C_1, C_2\}\big)\big) \tag{25}$$

Furthermore, if $\mathcal{H}_0^{\{C_1,C_2\}}$ *is true, then*

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(p(\mathbf{x}) \leq \alpha \mid C_1, C_2 \in \mathcal{C}(\mathbf{X})\Big) = \alpha, \quad \forall\, 0 \leq \alpha \leq 1. \tag{26}$$

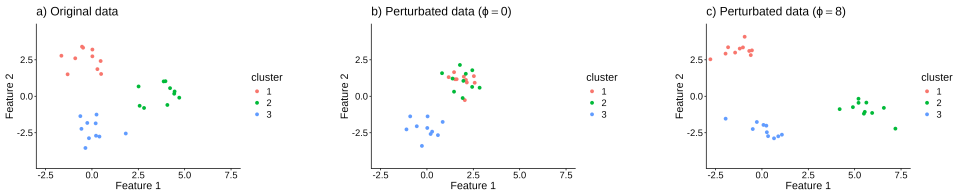
Note that I have added spacing and in the previous equations. The control of the selective type I error is proved in Appendix A. Here, you should add a paragraph describing what does the set $\mathcal{S}$ represent (explaining that it correspond to the set of perturbations that preserve the clustering, indicating how that perturbation is defined. You can refer to Section 2.2 in [4]. Then you should mention that the authors characterize $\mathcal{S}$ for several HAC algorithms so that it can be efficiently computed.

After a few manipulations of the definitions of dir and $\pi_\nu^\perp$, we obtain the following expression for $\mathbf{x}'(\phi)$:

$$\begin{aligned} \mathbf{x}'(\phi) &= \pi_\nu^\perp \mathbf{x} + \frac{\nu}{||\nu||_2} \operatorname{dir}(\nu^\top \mathbf{x})\phi \\ &= \mathbf{x} - \frac{\nu\nu^\top}{||\nu||_2^2}\mathbf{x} + \frac{\nu}{||\nu||_2} \operatorname{dir}(\nu^\top \mathbf{x})\phi \\ &= \mathbf{x} + \frac{\nu}{||\nu||_2^2}(\phi \operatorname{dir}(\nu^\top \mathbf{x}) - \nu^\top \mathbf{x}) \\ &= \mathbf{x} + \frac{\nu}{||\nu||_2^2}(\phi \operatorname{dir}(\nu^\top \mathbf{x}) - ||\nu^\top \mathbf{x}||_2 \operatorname{dir}(\nu^\top \mathbf{x})) \\ &= \mathbf{x} + \frac{\nu}{||\nu||_2^2} \operatorname{dir}(\nu^\top \mathbf{x})(\phi - ||\nu^\top \mathbf{x}||_2) \end{aligned} \tag{27}$$

$\mathbf{x}'(\phi)$ can be interpreted as a perturbed version of $\mathbf{x}$. Thus, $\mathcal{S}$ corresponds to the set of perturbations that preserve the clustering.

TODO: refaire fig sans (a), (b), (c) ...



(a) (b) (c)

Figure 3: (a) Original dataset $\mathbf{x}'(\phi) = \mathbf{x}$ with $\phi \approx 4.4$, (b) perturbated dataset $\mathbf{x}'(\phi)$ with $\phi = 0$ and (c) a perturbated dataset $\mathbf{x}'(\phi)$ with $\phi = 8$.

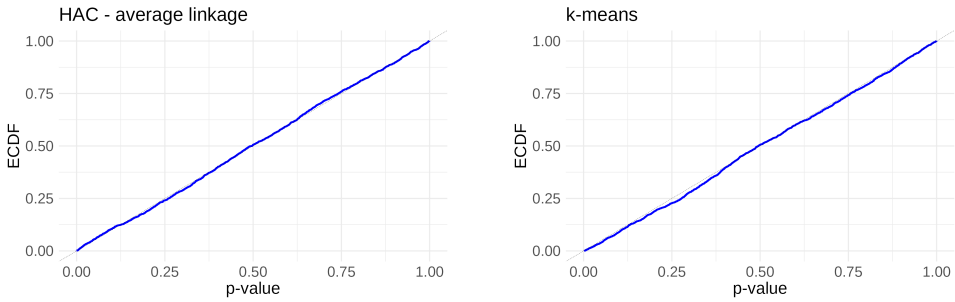
In order to compute the p-value (20) we have to characterize $\mathcal{S}\big(\mathbf{x}, \{C_1, C_2\}\big)$. Gao et al. propose algorithms for several HAC algorithms.

3.2 Numerical experiments

3.2.1 Uniformly distributed p-values under the null hypothesis

Section 3.1 detailed the approach of Gao *et al.* for testing the equality of cluster means after implementing a hierarchical agglomerative clustering (HAC) algorithm. Theorem 1 ensures that the p-value defined in (20) controls the selective type I error for clustering.

To empirically illustrate this result, we performed numerical simulations of the null distribution of p-values in the same setting as in Section 2.2.3, changing only the p-value. Only the test statistic? I believe you changed the entire p-value, given its form (24). Also, maybe you should recall here that we are simulating the null distribution of p-values, so that the reader doesn't have to go back to Section 2.2.3 if they don't want to know the details but just understand what you are doing here. For the simulations based on the HAC algorithm, we used the `clusterpval` library developed by Gao *et al.*, which implements the p-value (24). For the simulations based on the k-means algorithm, we used the `KmeansInference` library [2] Here, cite the corresponding paper: https://www.jmlr.org/papers/v24/22-0371.html, which relies on a similar p-value adapted to the k-means procedure. The corresponding results are presented in Figure 4, which shows that the resulting p-values are indeed super-uniform under the null hypothesis.



(a) (b)

Figure 4: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for (a) hierarchical agglomerative clustering (HAC) and (b) k-means algorithms. The distribution functions were computed using $M = 2000$ simulations from a multivariate centered normal distribution, as detailed in Section 2.2.3.

3.2.2 Robustness to feature independence

The control of the selective type I error for clustering relies on a strong assumption: the features I believe you want to refer to the *features*, not the observations. must be independent 13. You have already added the following equation before in the text: (13). Don't write it twice and say instead: (recall Equation 13).

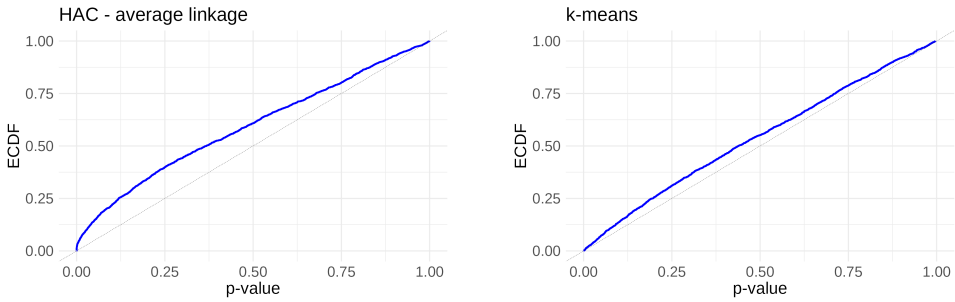
To determine whether the method is robust to this assumption, we simulated, in the same setting as in Section 3.2.1, centered Gaussian random vectors:Write the model instead, using the matrix-normal formulation and specifying the covariance matrix that you used.

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbb{I}_n, \Sigma) \tag{28}$$

with

$$\Sigma = \rho^{|i-j|} \text{ for } 1 \leq i, j \leq p \text{ and } \rho \in (0, 1].$$

what simulation are you performing? Specify that you are repeating the same simulation as in the previous section. Figure 5 shows that the p-values are no longer super-uniform under the null hypothesis, and therefore the selective type I error is no longer controlled when the independence assumption is not satisfied.

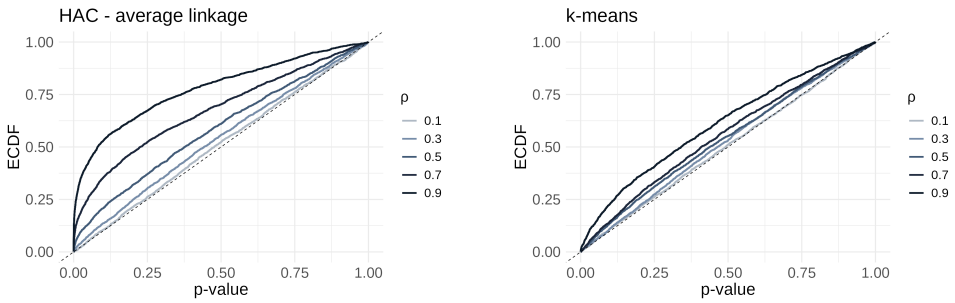


(a) (b)

Figure 5: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for (a) hierarchical agglomerative clustering (HAC) and (b) k-means algorithms. The distribution functions were computed using $M = 2000$ simulations from a gaussian random vector with a correlation matrix Σ and $\rho = \frac{1}{2}$.

The previous simulation is great. An idea if you want to: if you are simulating vectors with covariance $\rho^{|i-j|}$, simulate for various $\rho \in [0, 1)$ to illustrate how the p-values deviate from uniformity when ρ increases from 0 to 1. Then you can add all the curves for a set of ρ values in the same plot, colouring them with a gradient scale as a function of ρ.

By varying the value of ρ, we can observe the tendency of the p-values to deviate from uniformity. Figure 6 shows that the stronger the dependence between the features, the less the selective type I error is controlled.

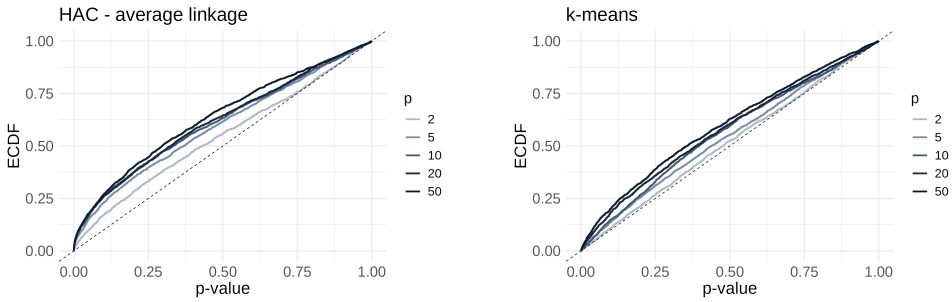


(a) (b)

Figure 6: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for hierarchical agglomerative clustering (HAC). The distribution functions were computed using $M = 2000$ simulations from a gaussian random vector with a correlation matrix Σ for different values of ρ.

3.2.3 Curse of dimensionality

The deviation from super-uniformity also increases with the dimension, as shown in Figure 7.



(a) (b)

Figure 7: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for hierarchical agglomerative clustering (HAC). The distribution functions were computed using $M = 2000$ simulations from a gaussian random vector with a correlation matrix Σ with $\rho = \frac{1}{2}$ for different values of p.

3.3 Data applications

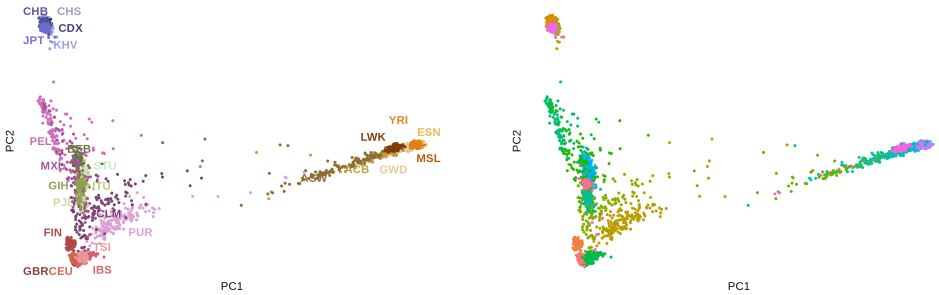
In this section, we analyze the `1KGP_100PC` dataset from the 1000 Genomes Project.

3.3.1 Data

The DNA sequence contains all the genetic information associated with an individual. It is composed of two complementary strands made of nucleotides, denoted A, C, G, and T. Although these sequences are very long, the vast majority of the genome is shared among all individuals. A reference genome has been defined, and genetic databases are constructed by counting nucleotide differences between each individual and this reference genome. From this matrix of genetic variations, a principal component analysis (PCA) can be performed in order to obtain the dataset `1KGP_100PC`.

3.3.2 Clustering

To try to capture the population structure of the indivuduals, we perfomed clustering with the same number of clusters as there are populations. Figure 8 shows the individuals represented on the first two principal components (PCs).



(a) (b)

Figure 8: Projection of the individuals onto the first two PCs, colored by population in (a) and by clusters obtained from HAC on the first 100 PCs in (b).

The clusters obtained with the HAC algorithm in Figure 8 (b) appear to be largely consistent with the population labels of the individuals. This can also be assessed by computing, for each cluster and each population, the corresponding proportion of individuals, as shown in Figure 9. Overall, the heatmap exhibits an approximately diagonal structure, indicating that each cluster mostly corresponds to a single population. However, some populations are spread across several clusters (PJL, ASW, GIH), and some clusters contain individuals from multiple populations (1, 3, 16).

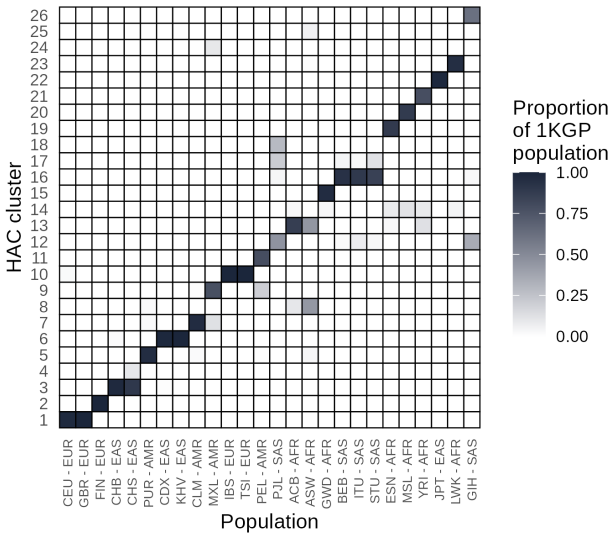
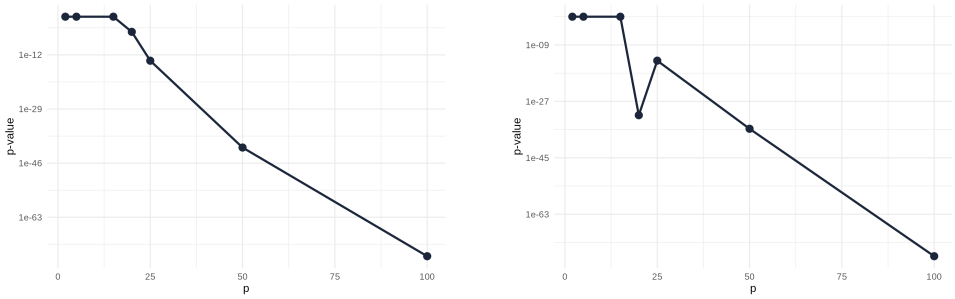


Figure 9: Heatmap showing the proportion of individuals from each population within each cluster obtained using the HAC algorithm with Euclidean distance and the Ward.D linkage method.

3.3.3 Curse of dimensionality

In Section 3.2.3, we showed that the p-value is sensitive to dimensionality. It is desirable to retain the maximum number of principal components to minimize information loss. However, this comes at a cost to the statistical guarantees of Gao's method. Indeed, even under $\mathcal{H}_0$, the p-value decreases as dimensionality increases. In Figure 10, we perform a test of equality of means between two clusters containing individuals from the same population.

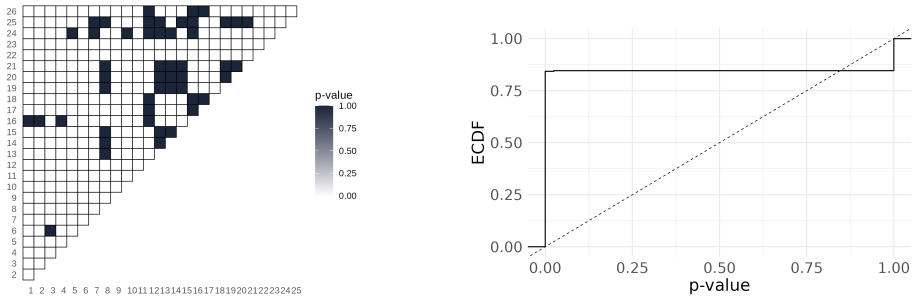


(a) (b)

Figure 10: p-values obtained from the test of equality between clusters 12 and 26 (a) and between clusters 8 and 13 (b), as a function of the number of PCs used for clustering

3.3.4 Intercontinental vs intracontinental

Each population belongs to a super-population, corresponding to a larger grouping similar to a continent. For example, Mexican Ancestry in Los Angeles, California (MXL) and Puerto Rican in Puerto Rico (PUR) belong to the AMR super-population, whereas Utah residents with Northern and Western European ancestry (CEU) belongs to the EUR super-population. Intuitively, the p-values associated with tests between clusters containing individuals from different continents are expected to be close to 0, while the p-values associated with tests between clusters containing individuals from the same continent are expected to be close to 1. To compute the p-values, we kept only the first 15 principal components. The resulting p-values are shown in Figure 11.

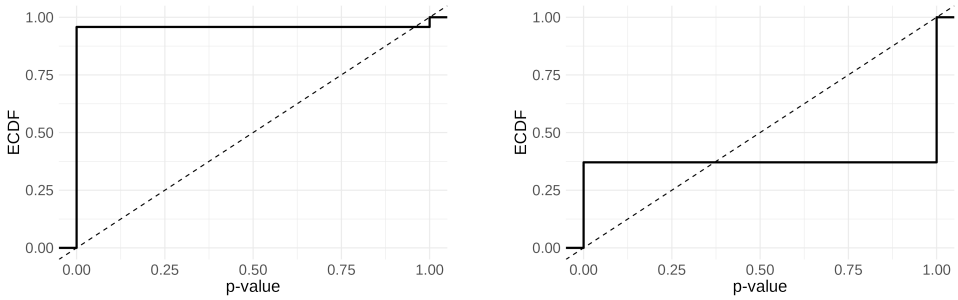


(a) (b)

Figure 11: (a) Heatmap of the p-values obtained using the `test_hier_clusters_exact` function applied to the first 15 PCs, with clusters obtained from HAC using Euclidean distance and the Ward.D linkage method on the first 100 principal components. (b) Empirical cumulative distribution function of the corresponding p-values.

la ECDF montre clairement que la pval n'est pas surper uniform... ie en pratique pval de Gao pas très exploitable ? il faut que je choissise encore moins de PCs ??

The Figure 12 shows that the p-values from the tests of equality of cluster means for different super-populations are concentrated near 0, whereas those from tests of clusters within the same superpopulation are mostly equal to 1.



(a) (b)

Figure 12: ECDF of p-values obtained from the test of equality between clusters means for pairs of clusters within the same super-population (a) and for pairs of clusters from different super-population (b)

A Proofs

Proof of Proposition 1. Let G_X denote the generalized inverse (quantile function) of F_X, defined for $u \in [0,1]$ as

$$G_X(u) = \inf\{x \in \mathbb{R} : F_X(x) \geq u\}. \tag{29}$$

By definition of the generalized inverse,

$$\{F_X(Y) \leq u\} = \{Y < G_X(u)\}. \tag{30}$$

Therefore,

$$\mathbb{P}\big(F_X(Y) \leq u\big) = \mathbb{P}\big(Y < G_X(u)\big) = F_Y\big(G_X(u)^-\big), \tag{31}$$

where $G_X(u)^-$ denotes the left limit at $G_X(u)$.

Since $X \preceq_{\text{st}} Y$, we have $F_Y \leq F_X$ pointwise, and thus

$$F_Y\big(G_X(u)^-\big) \leq F_X\big(G_X(u)^-\big). \tag{32}$$

By the defining property of the generalized inverse,

$$F_X\big(G_X(u)^-\big) \leq u. \tag{33}$$

Combining these inequalities yields

$$\mathbb{P}\big(F_X(Y) \leq u\big) \leq u, \quad \forall\, u \in [0,1], \tag{34}$$

which proves that $F_X(Y)$ is super-uniform.

□

Proof of Remark 1. By the proposition 1, we have that $F_X(X) \sim \mathcal{SU}(0,1)$. If F_X is continuous, then $F_X(G_X(u)) = u$ for all $u \in [0,1]$, and hence

$$\mathbb{P}\big(F_X(X) \leq u\big) = F_X\big(G_X(u)\big) = u, \quad \forall\, u \in [0,1], \tag{35}$$

which concludes the proof.

□

Proof of Proposition 2. By definition of the rejection rule,

$$\mathbb{P}_{\mathcal{H}_0}(\text{Reject } \mathcal{H}_0) = \mathbb{P}_{\mathcal{H}_0}(p \leq \alpha). \tag{36}$$

Since p is a p-value for $\mathcal{H}_0$, it is super-uniform under $\mathcal{H}_0$, hence

$$\mathbb{P}_{\mathcal{H}_0}(p \leq \alpha) \leq \alpha. \tag{37}$$

This establishes control of the type I error at level α. □

Proof of Proposition 3. Let $F_{T'(X)}$ denote the distribution function of $T'(X)$ under $\mathcal{H}_0$. By Proposition 1, the stochastic dominance $T'(X) \preceq_{\text{st}} T(X)$ implies that

$$F_{T'(X)}\big(T(X)\big) \sim \mathcal{SU}(0,1).$$

By definition,

$$p(x) = \mathbb{P}_{\mathcal{H}_0}\big(T'(X) \geq T(x)\big) = 1 - F_{T'(X)}\big(T(x)\big). \tag{38}$$

Let $u \in [0,1]$. Then

$$\begin{aligned}
\mathbb{P}_{\mathcal{H}_0}\big(p(X) \leq u\big) &= \mathbb{P}_{\mathcal{H}_0}\big(1 - F_{T'(X)}(T(X)) \leq u\big) && (39)\\
&= \mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \geq 1 - u\big) && (40)\\
&= 1 - \mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \leq 1 - u\big)\,. && (41)
\end{aligned}$$

Since $F_{T'(X)}(T(X))$ is super-uniform,

$$\mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \leq 1 - u\big) \leq 1 - u, \tag{42}$$

and therefore

$$\mathbb{P}_{\mathcal{H}_0}\big(p(X) \leq u\big) \leq u. \tag{43}$$

Thus, p is super-uniform under $\mathcal{H}_0$, and hence a p-value. □

Proof of Remark 2. When $T' = T$

$$p(X) = \mathbb{P}_{\mathcal{H}_0}\big(T(X') \geq T(X)\big) = 1 - F_T\big(T(X)\big), \tag{44}$$

By Remark 1, if F_T is continuous,

$$F_T\big(T(X)\big) \sim \mathcal{U}(0,1).$$

Consequently, for any $u \in [0,1]$,

$$\begin{aligned}
\mathbb{P}_{\mathcal{H}_0}\big(p(X) \leq u\big) &= \mathbb{P}_{\mathcal{H}_0}\big(1 - F_T(T(X)) \leq u\big) && (45)\\
&= \mathbb{P}_{\mathcal{H}_0}\big(F_T(T(X)) \geq 1 - u\big) && (46)\\
&= 1 - \mathbb{P}_{\mathcal{H}_0}\big(F_T(T(X)) \leq 1 - u\big) && (47)\\
&= 1 - (1 - u) && (48)\\
&= u. && (49)
\end{aligned}$$

Thus p is uniformly distributed on $[0,1]$ under $\mathcal{H}_0$. □

A.1 Proofs of Section 3

Proof of Lemma 1. i)

$$\begin{aligned}\mathbb{E}(\nu^\top \mathbf{X}(\pi_\nu^\perp \mathbf{X})^T) &= \mathbb{E}(\nu^\top \mathbf{X}\mathbf{X}^\top (\pi_\nu^\perp)^\top) \\ &= \nu^\top \mathbb{E}(\mathbf{X}\mathbf{X}^\top)(\pi_\nu^\perp)^\top \\ &= \nu^\top (\mathbb{V}(\mathbf{X}) + \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top)(\pi_\nu^\perp)^\top.\end{aligned} \tag{50}$$

Let $X \sim \mathcal{MN}(\mu, \mathbf{U}, \mathbf{\Sigma})$. Then, let us state $\mathbb{V}(\mathbf{X}) = \mathbf{U}\,\mathrm{Trace}(\mathbf{\Sigma})$. We prove the previous equality as a technical lemma after the proof.

In this case $\mathbb{V}(\mathbf{X}) = \mathbb{I}_n\,\mathrm{Trace}(\sigma^2 \mathbb{I}_p) = \sigma^2 p \mathbb{I}_n$

So

$$\begin{aligned}\mathbb{E}(\nu^\top \mathbf{X}(\pi_\nu^\perp \mathbf{X})^T) &= \nu^\top (\sigma^2 p \mathbb{I}_n + \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top)(\pi_\nu^\perp)^\top \\ &= \sigma^2 p \nu^\top (\pi_\nu^\perp)^\top + \nu^\top \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top (\pi_\nu^\perp)^\top \\ &= \sigma^2 p \nu^\top (\mathbb{I}_n - \frac{\nu\nu^\top}{||\nu||_2^2}) + \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}(\pi_\nu^\perp \mathbf{X})^\top \\ &= \sigma^2 p (\nu^\top - \frac{||\nu||_2^2 \nu^\top}{||\nu||_2^2}) + \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}((\pi_\nu^\perp \mathbf{X})^\top) \\ &= \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}((\pi_\nu^\perp \mathbf{X})^\top).\end{aligned} \tag{51}$$

ii)

$$\nu^T \mathbf{X} \sim \mathcal{MN}_{1\times p}(\nu^\top \mu, \nu\nu^\top, \sigma^2 \mathbb{I}_p) = \mathcal{N}_p(\nu^\top \mu, ||\nu||_2^2 \sigma^2 \mathbb{I}_p). \tag{52}$$

So $\nu^T \mathbf{X} \sim \mathcal{N}_p(0, ||\nu||_2^2 \sigma^2 \mathbb{I}_p)$ under $\mathcal{H}_0$. Yet, the density of a centered Gaussian random vector depends only on its norm, due to symmetry with respect to the origin. So $||\nu^\top \mathbf{X}||$ and $\mathrm{dir}(\nu^\top \mathbf{X})$ are independent.

□

Lemma 2.

Proof of Lemma 2.

$$\begin{aligned}\text{Let } \mathbf{Y} &= \mathbf{X} - \mu \\ \mathbb{V}(\mathbf{X}) &= \mathbb{E}[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^\top] \\ &= \mathbb{E}(\mathbf{Y}\mathbf{Y}^\top).\end{aligned} \tag{53}$$

And

$$\begin{aligned}(\mathbf{Y}\mathbf{Y}^T)_{ij} &= \sum_{k=1}^{p} \mathbf{Y}_{ik}\mathbf{Y}_{kj}^\top \\ &= \sum_{k=1}^{p} \mathbf{Y}_{ik}\mathbf{Y}_{jk}.\end{aligned} \tag{54}$$

Therefore,

$$\mathbb{E}((\mathbf{Y}\mathbf{Y}^\top)_{ij}) = \sum_{k=1}^{p} \mathbb{E}(\mathbf{Y}_{ik}\mathbf{Y}_{jk}). \tag{55}$$

As $\mathbf{Y}$ is centered, by definition of a normal matrix we have:

$$\begin{aligned}\mathbb{E}((\mathbf{Y}\mathbf{Y}^\top)_{ij}) &= \sum_{k=1}^{p} \mathbf{U}_{ij}\Sigma_{kk} \\ &= \mathbf{U}_{ij} \sum_{k=1}^{p} \Sigma_{kk} \\ &= \mathbf{U}_{ij}\,\mathrm{Trace}(\Sigma).\end{aligned} \tag{56}$$

□

Proof of Equation 26 in Theorem 1. From the definition (20), we have that for all $\mathbf{x} \in \mathbb{R}^{n\times p}$

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left(p(\mathbf{x}) \le \alpha \mid C_1, C_2 \in \mathcal{C}(\mathbf{X}), \pi_\nu^\perp \mathbf{X} = \pi_\nu^\perp \mathbf{x}, \mathrm{dir}(\nu^\top \mathbf{X}) = \mathrm{dir}(\nu^\top \mathbf{x})\right) = \alpha. \tag{57}$$

Therefore, we have

$$\begin{aligned}&\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left(p(\mathbf{x}) \le \alpha \,\middle|\, C_1, C_2 \in \mathcal{C}(\mathbf{X})\right) \\ &= \mathbb{E}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left[\mathbb{1}\{p(\mathbf{x}) \le \alpha\} \,\middle|\, C_1, C_2 \in \mathcal{C}(\mathbf{X})\right] \\ &= \mathbb{E}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left[\mathbb{E}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left[\mathbb{1}\{p(\mathbf{x}) \le \alpha\} \,\middle|\, C_1, C_2 \in \mathcal{C}(\mathbf{X}), \pi_\nu^\perp \mathbf{X} = \pi_\nu^\perp \mathbf{x},\right.\right. \\ &\qquad\qquad \left.\left.\mathrm{dir}(\nu^\top \mathbf{X}) = \mathrm{dir}(\nu^\top \mathbf{x})\right] \,\middle|\, C_1, C_2 \in \mathcal{C}(\mathbf{X})\right] \\ &= \mathbb{E}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left[\alpha \,\middle|\, C_1, C_2 \in \mathcal{C}(\mathbf{X})\right] \\ &= \alpha,\end{aligned} \tag{58}$$

where the second equality follows from the law of total expectation. □

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